Practice Questions for 2024-10-22-Convexity

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1 Questions

1. According to Definition 1, which of the following statements is true about a convex set S in R^m?

A. For any two points z_1 and z_2 in S, the line segment connecting them, defined by $tz_1 + (1-t)z_2$ where $t \geq 0$, is entirely contained within S.

B. For any two points z_1 and z_2 in S, the line segment connecting them, defined by $tz_1 + (1-t)z_2$ where $0 < t < 1$, is entirely contained within S.

C. For any two points z_1 and z_2 in S, the line segment connecting them, defined by $tz_1 + (1-t)z_2$ where $t \leq 0$, is entirely contained within S.

D. For any two points z_1 and z_2 in S, the line segment connecting them, defined by $tz_1 + (1-t)z_2$ where $t = 1$, is entirely contained within S.

E. For any two points z_1 and z_2 in S, the line segment connecting them, defined by $tz_1 + (1-t)z_2$ where t is any real number, is entirely contained within S.

2. According to Theorem 10.1, which of the following is a necessary and sufficient condition for a set S to be convex?

A. S contains at least one convex combination of its points.

B. S contains all convex combinations of its points.

C. S contains no convex combinations of its points.

D. S contains only the endpoints of convex combinations of its points.

E. S contains only a subset of convex combinations of its points.

3. In the proof of Theorem 10.1, when demonstrating that if S is convex, then it contains all convex combinations of its points, how is the case for n=3 handled?

A. It is shown that a convex combination of three points can be expressed as a convex combination of two points.

B. It is shown that a convex combination of three points can be expressed as a convex combination of a convex combination of two points and a third point.

C. It is shown that a convex combination of three points can be expressed as a linear combination of the three points.

D. It is shown that a convex combination of three points can be expressed as a convex combination of four points.

E. It is shown that a convex combination of three points can be expressed as a single point.

2 Answers

1. According to Definition 1, which of the following statements is true about a convex set S in R^m?

   A. Answer A is incorrect because it includes $t = 0$ and $t = 1$, which are the endpoints of the line segment, not the line segment itself.

   B. Answer B is correct because it accurately reflects the definition of a convex set, where the line segment connecting any two points within the set must also be within the set, with $0 < t < 1$ defining the line segment.

   C. Answer C is incorrect because it uses $t \leq 0$, which is not the correct range for the parameter t in the definition of a convex set.

   D. Answer D is incorrect because it only considers the case where $t = 1$, which is just one of the endpoints of the line segment, not the line segment itself.

   E. Answer E is incorrect because it allows t to be any real number, which does not define the line segment between z_1 and z_2.

2. According to Theorem 10.1, which of the following is a necessary and sufficient condition for a set S to be convex?

   A. Answer A is incorrect because the theorem requires that S contains *all* convex combinations, not just one.

   B. Answer B is correct because Theorem 10.1 states that a set S is convex if and only if it contains all convex combinations of its points.

   C. Answer C is incorrect because a convex set must contain convex combinations of its points.

   D. Answer D is incorrect because a convex set must contain the entire convex combination, not just the endpoints.

   E. Answer E is incorrect because a convex set must contain *all* convex combinations, not just a subset.

3. In the proof of Theorem 10.1, when demonstrating that if S is convex, then it contains all convex combinations of its points, how is the case for n=3 handled?

   A. Answer A is incorrect because the proof does not directly reduce the convex combination of three points to a convex combination of two points, but rather uses a nested approach.

   B. Answer B is correct because the proof for n=3 rewrites the convex combination of three points as a convex combination of a convex combination of two points and the third point, leveraging the base case (n=2).

   C. Answer C is incorrect because the proof specifically uses convex combinations, not just any linear combination.

   D. Answer D is incorrect because the proof does not involve a convex combination of four points.

   E. Answer E is incorrect because a convex combination of three points is not generally a single point.